Seat: TBD PETER ANTEATER PANTEATE

# CompSci 161 Diagnostic Exam 1, Fall 2025

# DO NOT OPEN EXAM UNTIL INSTRUCTED TO DO SO

# SILENCE AND STORE ALL ELECTRONICS

This is a diagnostic exam intended to help you evaluate your readiness for the real exam.

The following rules apply to you, whether you think they do or not. Read and understand them; failure to abide by these rules, or directions given by course staff during the exam, may result in disciplinary action, including but not limited to a failing grade in the class.

- This exam is solely for students enrolled in this lecture. Anyone not enrolled in this lecture may not take an exam.
- You may not open the exam or begin writing until the instructor has explicitly given you permission to do so.
- Keep acceptable identification readily accessible during the test. Proctors may request to see it.
- This exam is closed book, closed notes, and is individual effort. Once course staff begin passing out exams, you may not communicate with anyone other than proctors for any reason, nor may you have electronics, including calculators watches and phones, available to you during the test for any reason. **YOU DO NOT NEED A CALCULATOR!**
- If you leave your seat during the test for any reason, your instructor may collect it and deem you to have turned it in. Do not ask proctors for an exemption to this, they are not authorized to grant such.
- You must take the exam in your assigned seat unless the professor (not a TA) tells you otherwise. The instructor will call to cease writing at 6:45 PM, at which point you must immediately cease writing and close the exam. You may not write any further at that point, including finishing one's current sentence.
- If you believe a question is ambiguous, write at least two reasonable interpretations and indicate clearly which one you will be using. Then answer your question with that assumption. Unless your interpretation makes the problem much more trivial than intended, we will grade your response as if one of us had made that clarification.
- The purpose of the real exam is to evaluate how well *you* understand the material presented in the course. It is an academic integrity violation to do anything that subverts the goals of this assessment including, but not limited to, not doing your own work or submitting that of anyone else.
- We will only grade responses marked in the space provided for each question.

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**Nothing you write on this page will be graded.** The next page in this booklet contains a spot to answer these questions. You may use this page as scratch paper if you would like, and room to do so exists.

1. (2 points) Suppose we are going to be sorting a vector using quickSort, the relevant code of which is below. When we choose a pivot in the partition step, we choose uniformly at random among the possible ones.

   ```
   QuickSort(A, start, end)
     if start < end then
       q = partition(A, start, end)
       QuickSort(A, start, q − 1)
       QuickSort(A, q + 1, end)
   ```

   The array, after pivoting using key 67, is:

   | | | | | | | | | | | | | | | | |
   |---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
   | 40 | 25 | 52 | 17 | 30 | 48 | 26 | 65 | 46 | 58 | 33 | 67 | 68 | 89 | 81 | 72 |

   We will then finish quickSort, using "pure" quickSort until the array is sorted. For each of the following pairs of keys, will they be compared during the run of the program? If they will definitely be compared, answer "yes" for that pair. If they will definitely not be compared, answer "no." If they might be, but are not guaranteed to be, answer "maybe."

   (a) 17 and 40 (b) 48 and 52 (c) 58 and 68

2. (2 points) Consider the following vector:

   | | | | | | | | | | |
   |---|---|---|---|---|---|---|---|---|---|
   | 978 | 167 | 103 | 386 | 987 | 335 | 448 | 298 | 582 | 215 |
   | 842 | 640 | 867 | 943 | 998 | 784 | 598 | 739 | 145 | 595 |
   | 384 | 594 | 966 | 724 | 231 | 948 | 163 | 578 | 903 | 748 |
   | 794 | 928 | 663 | 702 | 220 | 556 | 937 | 569 | 659 | 520 |
   | 589 | 502 | 965 | 457 | 351 | | | | | |

   This vector is represented here with elements 1-10 in the first row, 11-20 in the second row, 21-30 in the third row, 31-40 in the fourth row, and 41-45 in the last row.

   Suppose we run the deterministic selection algorithm from lecture. What is the first pivot chosen for the first partition step? Write *only* the integer value.

Recall the Master Theorem:

$T(n) = aT(n/b) + f(n)$, for some $a \geq 1$, $b > 1$, and $f(n)$ is asymptotically positive.

- If there is a small constant $\varepsilon > 0$ such that $f(n)$ is $\mathcal{O}(n^{\log_b a - \varepsilon})$, then $T(n)$ is $\Theta(n^{\log_b a})$
- If there is a constant $k \geq 0$, such that $f(n)$ is $\Theta(n^{\log_b a} \log^k n)$, then $T(n)$ is $\Theta(n^{\log_b a} \log^{k+1} n)$
- If there is a small constant $\varepsilon > 0$ such that $f(n)$ is $\Omega(n^{\log_b a + \varepsilon})$, then $T(n)$ is $\Theta(f(n))$.

3. (2 points) Express all three of the following recurrences in closed form.

   - $T_1(n) = 3T_1(n/3) + 15$
   - $T_2(n) = 3T_2(n/3) + 15n$
   - $T_3(n) = 3T_3(n/3) + n^2$

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For question 1, circle the correct answer for each part.

($a$) yes no maybe

($b$) yes no maybe

($c$) yes no maybe

Write your answers to 2 and 3 in the relevant boxes.

| $2a$ | $3 : T_1(n)$ | $3 : T_2(n)$ | $3 : T_3(n)$ |
|---|---|---|---|
| | | | |

You may use the rest of this page as scratch paper, but nothing you write under the dotted line will be graded.

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**Nothing you write on this page will be graded.** The next page in this booklet contains a spot to answer these questions. You may use this page as scratch paper if you would like, and room to do so exists.

4. (2 points) Suppose we have a directed graph $G = (V, E)$ where each edge is assigned a color from {red, white, blue}. A walk in $G$ is called *patriotic* if its sequence of edge colors is red, white, blue, red, white, blue, and so on. More formally, a walk $v_0 \to v_1 \to \ldots \to v_k$ is *patriotic* if, for every integer $i$, the edge $v_i \to v_{i+1}$ is red if $i \equiv 0 \pmod 3$, white if $i \equiv 1 \pmod 3$, and blue if $i \equiv 2 \pmod 3$.

   Use graph modeling to write an algorithm to find all vertices in $G$ that can be reached from a designated vertex $s$ via a patriotic walk. Give and briefly justify the running time of your approach. For full credit, your approach should have linear running time. You may make any reasonable assumption about how the graph is represented that you would like; if it affects your algorithm, write the assumption in the answer space.

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Write your answer to question 4 on this page. Note that you might not use the full space for your answer, and that is okay.

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5. (1 point) Recently, I was looking back at my career as a professor and I was thinking of some of my former students, each of whom has already graduated and moved onto a great career, as I hope all of you will soon (I will miss you though). I was wondering which pair of students were *both* students concurrently for the longest period of time. To answer this question, I had to gather some information. From the registrar, I was able to get very precise enrollment and graduation dates for each. The registrar, for purposes of this question anyway, defines a student's **enrollment** time as the precise millisecond that the student's intent to register is recorded, and defines the student's **graduation** time as the precise millisecond that the student's diploma is printed. The time from enrollment to graduation is an individual's time as a student. Note that the definitions for enrollment and graduation in this problem mean that no two students have the exact same enrollment time, nor do any two have the exact same graduation time, even if they started the same quarter and ended the same quarter.

   Suppose you are given a list of $n$ students, each with the enrollment and graduation times as described above. You may assume the list is sorted by enrollment date if you prefer. Design an efficient **divide and conquer** algorithm to determine which two individuals' time as students has the maximum overlap.

   More formally, you want to find the pair of indices $i \neq j$ such that the following value is maximized:

   $$\min(A[i].\text{graduation}, A[j].\text{graduation}) - \max(A[i].\text{enrollment}, A[j].\text{enrollment})$$

   No credit will be given for an algorithm with running time $\Omega(n^2)$.

   *Do not assume people graduate in the order they register. Do not assume everyone graduates in four years, or 12 quarters, or 45 months, or anything like that.*

**This question counts towards the Divide and Conquer portion of the Competency of Core Topics in your grade.** Or at least it would if this were the real test.

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Write your answer to question 5 on this page. Note that you might not use the full space for your answer, and that is okay.

Scratch page. Nothing you write on this side of the page will be graded.

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